

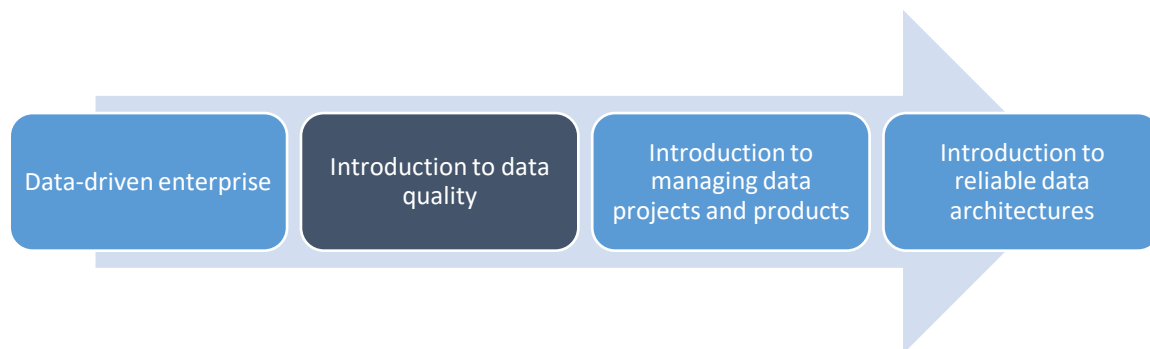
Topic 2- Introduction to Data Quality

This document is the handbook for Topic 2 – **Introduction to Data Quality** – within Module 1 – **Data Fundamentals**.

The purpose of this document is to guide your learning throughout this topic and help you to maximise the value you get from the materials provided by the BPP School of Technology.

Context

This handbook is for one of 4 topics for this Module.



Every topic contributes towards the ultimate learning objectives for the Module, which you will be assessed on at the end of the term.

Module Learning Outcomes

On successfully completing this module, you will be able to:

- **Recognise** fundamental data types and data source types, key Data Engineering technological standards and best practices, and relevant regulations
- **Apply** fundamental principles for fostering a data-driven culture through collaborating with diverse stakeholders in Agile and Lean teams, avoiding waste and respecting organisational processes for data quality and data project management

- **Model** standardised Big Data ecosystems and architectures for enterprise data using visual approaches and comprehend how to design rudimentary data products, and how they add value to the organisation
- **Explain** the importance of assessing, improving, and maintaining data quality using frameworks and methodologies to ensure data accuracy, completeness, and consistency.

Module Assessment

The Level 5 Data Engineer EPA has two assessment methods, each with its own mapping of KSBs. The Assessment plan and assessment guidance documents above list the criteria and KSBs that are assessed. The criteria group the KSBs and describe what the apprentice needs to do to achieve a pass or distinction for that assessment method.

Both assessment methods need to be passed by the candidate:

(1) Project with report

The learner will complete a project and write a report of 3500 words. Project brief submitted at gateway:

- Learners will have 10 weeks to complete the project and submit the report to the EPAO
- Learners also need to prepare and give a presentation to an independent assessor on their project
- The presentation with questions will last at least 50 minutes. The independent assessor will ask at least 6 questions about the project and presentation
- The project has to have real business application and benefit. Candidates are expected to showcase the use of appropriate standards for sustainability, privacy and security, thoroughly document their data pipeline designs, explain the choice of relevant tooling and demonstrate operational awareness of deployment, access control, risks, and how other stakeholders may be impacted positively and negatively

(2) Professional discussion underpinned by a portfolio of evidence

- Learners will have a professional discussion with an independent assessor. It will last 80 minutes.

- They will be asked at least 10 questions about Data Engineering.
- The portfolio of evidence will be used to help answer the questions.
- We expect the candidates to demonstrate examples of working with data teams on data projects and data products, showcase ideas for future-proofing data, be clear on applying problem-solving skills, show regulatory awareness, and sensitivity towards data quality, data governance and areas for continuous improvement, both personal and organisational.

Link to summative assessment:

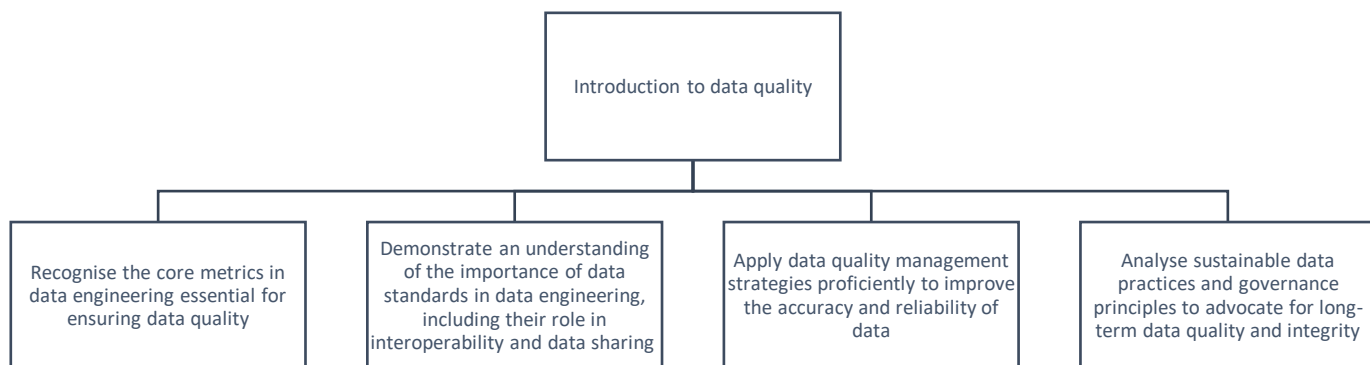
The "Introduction to Data Quality" topic is highly relevant to the Summative Assessment as covers key knowledge, skills, and behaviours required for both the project with report and the professional discussion components.

Specific areas of relevance include:

- Understanding data quality metrics for meeting technical and business requirements (relevant for the project)
- Importance of data standards, metadata, and governance for sustainability, privacy, and security (relevant for the project)
- Navigating quality issues in common data formats like XML, CSV, and JSON (relevant for documenting data pipelines and choosing tools in the project)
- Data quality management strategies like profiling, validation, and monitoring (relevant for the professional discussion)
- Data testing methodologies showcasing problem-solving skills and regulatory awareness (relevant for the professional discussion)
- Data policies and regulations like GDPR for responsible data handling (relevant for the professional discussion)
- Sustainable data practices and governance principles for future-proofing and continuous improvement (relevant for the professional discussion)

Topic Learning Outcomes

As a step towards build your skills towards the final module assessment, the learning objectives for this topic are:



Introduction

For data engineers, mastering data quality concepts and techniques is an indispensable skill. In today's data-rich world, organisations rely heavily on the accuracy, integrity, and reliability of their data assets to drive informed decision-making, optimise processes, and maintain a competitive edge. This topic is tailored to equip you with a comprehensive understanding of data quality principles and best practices, preparing them to tackle real-world challenges effectively.

Real-world examples of the value delivered by strong data quality practices abound. Consider the case of a leading retail company that faced significant challenges with inaccurate product data, leading to customer dissatisfaction and operational inefficiencies. By implementing a robust data quality testing framework, including data validation, continuous monitoring, and data profiling, the company successfully improved the accuracy and reliability of its product data, resulting in enhanced customer satisfaction and streamlined operations.

Another notable example is from the healthcare industry, where data quality is paramount for ensuring patient safety and coordinated care. By adhering to data quality principles, such as maintaining data lineage and implementing unique identifiers, healthcare organisations can trace the origin of patient records, ensuring compliance with privacy regulations and preventing errors like duplicate or misidentified records.

Structure

Topics for this programme follow a Prepare-Collaborate-Apply structure:

Prepare

This is the stage where you build the knowledge to underpin your learning. This might involve completing interactive e-learning packages, watching videos, or working through reading materials.

It is essential that you make the most of the learning materials provided before attending webinars, as this will allow you to test your knowledge and stretch your understanding further.

This e-learning for this topic covers a comprehensive range of topics related to data quality, equipping learners with the knowledge and skills essential for assessing and enhancing the accuracy, completeness, consistency, and timeliness of data.

Collaborate

This is where you interact with our expert tutors and coaches to shape and refine your understanding through discussion, testing and carrying out more advanced practical and realistic tasks. This also helps to develop valuable team-working skills.

The webinar for this topic will focus on data quality concepts, standards, handling issues, strategies, automation, sustainability considerations, and practical exercises related to data quality management for data engineers.

Apply

You now apply the knowledge you have developed to real-world tasks, including off-the-job (OTJ) and on-the-job (OJT).

Off-the-job learning tasks

This stage is all about ensuring you truly grasp and retain what you've learned.

Through completion of off-the-job (OTJ) revision tasks and tests, you'll get plenty of practice applying your knowledge. Plan to dedicate 6-8 hours each week to guided study and portfolio work, with sessions typically on the same day each week.

Task 1 brief: Exploring real-world datasets

The task aims to gain hands-on experience in exploring real-world datasets. For this task you should complete the following steps:

1. Find the "IBM HR Analytics Employee Attrition & Performance" dataset on Kaggle.
2. Explore the dataset's metadata, including assessing its usability score, credibility issues, number of columns, license, and usage rights.
3. Examine the dataset's description, data types (e.g., `sc:Integer`, `sc:Boolean`), and compression implications.
4. Download the dataset and identify potential data quality issues, such as column interpretation challenges, inconsistencies, and missing/invalid values.
5. Practice basic data cleaning and documentation tasks, including:
 - Retaining specific columns
 - Creating a "Metadata" sheet with document details
 - Adding a "Schema" sheet to define column properties
 - Adding a "Provenance" sheet to document the dataset's origin
6. Upload the cleaned and documented dataset to your GitHub repository.

7. Review the Knowledge, Skills, and Behaviours (KSBs) covered in this exercise and document them in your learning journal.

The goal is to develop skills in navigating data repositories, analysing metadata, identifying data quality issues, and performing basic cleaning and documentation tasks essential for data engineers.

Further guidance can be found here: [L5DE 1.2 Apply \(OTJ\)](#)

On-the-job learning tasks

This phase is where you'll put your knowledge into action and gain valuable hands-on experience in real-world scenarios. During the on-the-job (OJT) stages of your learning, you'll have the opportunity to apply the concepts you've learned in practical settings, refining your skills and enhancing your understanding.

Task 1 brief: Exploring automated data quality management tools

The task aims to gain hands-on experience with automated data quality management tools by researching, exploring, and documenting findings in your learning journal.

For this task you should:

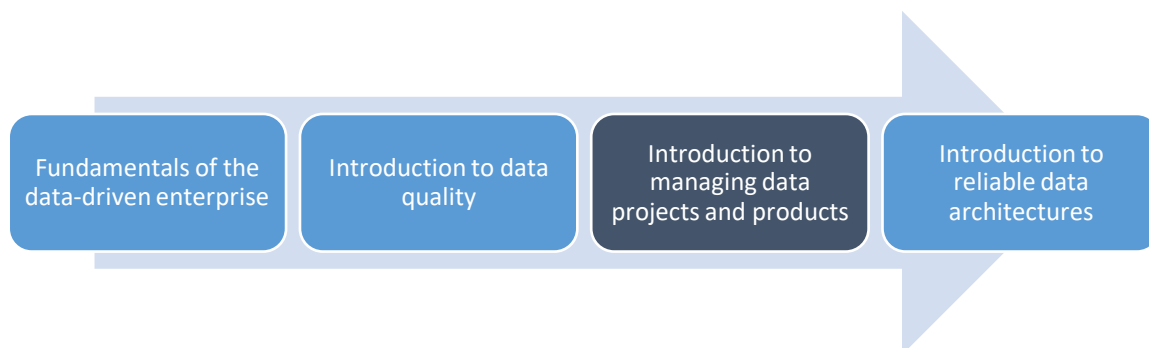
- Identify and research at least two automated data quality management tools or libraries used in your organization or relevant to your field of interest
- Arrange a shadowing session with a data engineer or subject matter expert to observe how they leverage these tools in their workflow. Document insights on use cases, benefits, challenges, integration, and best practices
- In your learning journal, provide an overview of the tools researched, key features and capabilities, examples of usage in your organisation/industry, potential limitations, and best practice recommendations
- Share your learning journal entry with your coach or mentor for feedback and guidance
- Log your on-the-job training hours using the provided file

The task emphasises practical exposure, documentation, and knowledge-sharing to enhance your understanding and skills in ensuring data quality and reliability using automated tools and techniques.

Further guidance can be found here: [L5DE 1.2 Apply \(OJT\)](#)

Link

This handbook is for one of 4 topics for this Module.



The sequence of topics in this module is carefully designed so that your knowledge and skills will develop as you progress.

The next topic is **Introduction to managing data projects and products**.